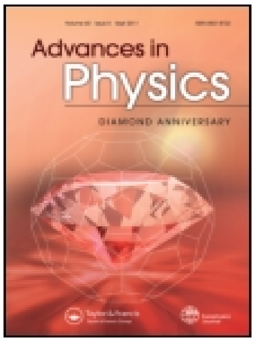

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Dynamics of wet granular matter

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Recent advances in understanding model systems of wet granular materials are presented, with particular emphasis on statistical concepts, dynamics, and phase transitions. It is demonstrated that although wet granular systems are quite complex, their main features may be understood on the basis of rather simple concepts. The significance of these systems for investigating fundamental problems of non-equilibrium dynamics are shortly discussed.

1. Introduction

Judging from the ease at which it quietly runs through the orifice of an hourglass, or forms wandering dunes in the wind like waves on the ocean, dry sand may be legitimately called a fluid. It will not maintain any shape imposed on it, and tends to form a horizontal boundary to the air above, very much resembling regular liquids, like water. However, when some of the latter is mixed into the dry sand, a pasty material emerges, which can be sculptured into quite stable structures, such as sand castles. It is this pasty state, resulting from the intimate mixture of a granulate with a certain amount of liquid, on which we will focus in this paper.

As an example, let us consider a land slide in wet soil, such as the one depicted as a bird's eye view in figure 1. First of all, we see that we are concerned with a threshold phenomenon: some part of the slope has massively moved downwards, whereas in the adjacent parts, where conditions were certainly similar, nothing happened at all. Furthermore, it is instructive to study the remains of the trail, which is clearly visible as a straight line to the left and right of the landslide region. On the slide region itself, it has been strongly deformed into a shape similar to a parabola. This shows that the slope has not come down as a more or less solid piece, sliding on a fault zone deeper inside the hill, but that it rather has liquefied, and crept down like a viscous fluid. Although some individual blocks of soil are still apparent, the overall liquid-like behavior is obvious [1]. It is tempting to suspect some kind of a discontinuous phase transition (from a solid to a liquid state) as the underlying phenomenon. These questions are far from settled presently [2].

Granular materials in general have concerned engineers for centuries, due to their vast occurrence in everyday life as well as in geology, mining, waste

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Figure 1. Massive landslide at La Conchita, California, which occurred in spring 1995 after heavy rain. Fortunately, no one was killed or injured (courtesy of U.S. Geological Survey [2]).

management, and many processes of industrial relevance. More than half of the raw materials world-wide come as solids, and thus in granular form. They must be stored in silos, conveyed to and from refinement plants, where they are mixed with other granulates or liquids, and so on. The necessity of quantitatively describing these processes, in order to design industrial plants or to build structures on various types of soil, has spurred an abundance of studies on the modelling and characterization of granular matter. The modelling of soil dynamics, for instance, is meanwhile a field of rather mature textbook knowledge [3–7]. However, the need for quantitative predictions has restricted these studies mainly to 'real' systems, which are far too complex for a microscopic approach. One rather uses continuum models, with empirical data of the material as an input [4–6]. A direct connection to (and hence an identification of) microscopic mechanisms was thus quite generally impossible. From the basic science point of view, one would rather try to find well-defined model systems which exhibit effects similar to those observed in the real systems. In other words: try to find the most simple system which is just complex enough to show the effect(s) of interest.

The discovery of granular matter as a matter of science is usually credited to R. A. Bagnold [8, 9]. Fascinated with the physical properties of sand dunes in the desert, he tried to derive a constitutive relation between the shear rate and the local granular temperature, in order to describe the observed avalanching behavior of dry sand. His results, which he obtained about fifty years ago, were much later corroborated rigorously by an analytical approach based on the Chapman–Enskog formalism [10]. Edwards and Oakeshott have come up with ideas for treating granular materials by means of statistical concepts [11]. However, only the advent of powerful computers has made possible the extensive study of the fundamental physics of these very complex systems, and has enabled reliable tests of theoretical models relating microscopic aspects to macroscopic properties. This is why the physical understanding of granular matter, in the sense of natural science, has seen significant advances in particular in the very recent past [12–25]. In order to get into the matter, it appears natural to start with a short overview of the physics of *dry* granulates, such as the sand we find in the desert, and to add effects due to moisture further below. For details, the reader is recommended to the excellent reviews which are available [16, 26–32].

1.1. *Dry granular systems*

There are three aspects which distinguish a dry granulate from a regular gas. First, it is obvious that the generally rough surfaces of the grains, be they of sand, ore, soil, food components, or other materials, are qualitatively different from the indistinguishable shapes of atoms or molecules. Strictly speaking, we are thus not dealing with an ensemble of identical particles, since each will have at least slightly different size and shape. It will therefore be important to pose questions to the system such as to make the answers as independent as possible from this inherent geometric poly-dispersity. In any case, it should be borne in mind that effects typical for identical particles, like crystallization, do not play any significant role in granulates. Instead, we are concerned throughout with random systems.

Second, in strong contrast to a molecular gas, the mass of each individual particle is many orders of magnitude larger than that of an atom. As a consequence, a grain is too heavy to be lifted noticeably from its support by thermal energies. If we take silicon oxide as an example (of which most sand is composed), we are to deal with a density $\varrho_{SiO} \approx 2.6\,\mathrm{g/cm^3}$. It is readily verified that for grains with a radius larger than 0.5 micron, thermal energies will not be sufficient to lift a grain by more than its own radius. Granular materials, however, are usually envisaged as being composed of grains with at least several microns in size. The direct result is that as long as it is not agitated mechanically, a granulate forms a condensed phase (with frozen-in disorder) at the bottom of its container, kept together by gravity alone.

The third, and very important point is that collisions between grains are noticeably inelastic. When two grains collide, some of the kinetic energy stored in the center of mass motions of the collision partners is transferred to their internal atomic degrees of freedom. As a result, their (relative) momentum after impact, p_f, will be less than that before impact, p_i. The ratio of these momenta (taken in the center of mass frame of reference) is called the restitution coefficient, $\epsilon := p_f/p_i$. This is an important quantity for the characterization of granular materials. In practice, it is usually well below 0.9 [32].

A very fundamental consequence of the last two aspects is seen if one interprets the kinetic energy of the random motion as a temperature. This can be defined as $T := m(\langle \mathbf{v}^2 \rangle - \langle \mathbf{v} \rangle^2)/3k_B$ [33], where m is the (average) mass of an individual grain, $\mathbf{v}$ its velocity vector, and k_B is Boltzmann's constant [34]. Even for only gently shaken sand, typical grain velocities are on the order of cm/sec. Assuming sufficient randomization of the motion, the relative velocities are of the same order of magnitude. Given the large mass of the grains mentioned above, one finds that the random granular motion corresponds to temperatures in the Terakelvin range. The dissipative nature of the inter-granular collisions intimately couples this Terakelvin heat bath to the atomic degrees of freedom within the grains, which are at room temperature. This extreme non-equilibrium situation gives rise to a large fraction of the striking properties of dry granular matter [29, 31, 32, 35].

With the granular temperature as defined above, the energy lost in a collision scales as $\Delta E \propto (1 - \epsilon^2)T$, and the collision frequency scales as $T^{1/2}/l$, where l is the mean free path (by referring to a well defined mean free path, we tacitly assume a uniform density of the system). We immediately get

$$\frac{dT}{dt} \propto -\frac{(1 - \epsilon^2)T^{3/2}}{l} \tag{1}$$

for the rate of cooling due to the inelastic collisions. This directly leads to Haff's law [36], which states that the granular temperature decays algebraically as $(1 + Ct)^{-2}$, where C is some constant. In contrast, if there is a constant shear rate, $\nabla \mathbf{v}$, imposed on the granulate, the latter is constantly heated by the extra relative motion of the grains. Sela and Goldhirsch have shown that this gives rise to an additional term $lT^{1/2}(\nabla \mathbf{v})^2$, such that the granular temperature obeys [10, 30]

$$\frac{dT}{dt} \propto lT^{1/2}(\nabla \mathbf{v})^2 - (1 - \epsilon^2)l^{-1}T^{3/2} \tag{2}$$

This seems to lead to an 'equilibrium' granular temperature, $T_0 = (l\nabla \mathbf{v})^2/(1 - \epsilon^2)$.

However, this is not a *stable* state [12, 37]. Consider a fluctuation corresponding to a locally increased density. Grains which enter this region will have more frequent impacts, and thus lose energy more rapidly than in regions of smaller density. The grains thus leave this region significantly slower than they entered it. Hence there will be a net flow of particles towards the high density region, and a clustering instability takes place [12, 16, 37]. This effect is particularly striking when observed in a container with several partitions which are connected by small holes. When the container is vibrated in order to maintain a certain granular temperature, and conditions are properly set, the grains will eventually go all in one of the partitions, leaving the other(s) empty [38–41]. At first glance, this seems to contradict the second law of thermodynamics and has led to the notion of a 'Maxwell demon' built into granular systems [38].

Another striking effect characteristic of granular systems is the so-called 'brazil nut effect' [42–48]. It refers, e.g., to the fact that in cereal packages containing nuts and smaller grains (e.g., oats) one usually observes that the nuts segregate to the surface of the granulate, although from buoyancy one would expect the opposite [49]. This is a rather general phenomenon among granular materials, which may be understood in a continuum picture as follows [46]. Consider a spherical region with radius R_{sph} within the granular gas. When a grain, with radius $R \ll R_{sph}$, enters this region, it will have many impacts and thereby lose a lot of energy before it can leave this region again. When the latter is replaced by a solid sphere (i.e., a large nut) with radius R_{sph}, the grain will experience only one such impact. Consequently, the granular gas is hotter close to an intruder than far away in the bulk, because the intruder is a less effective heat sink. The granular gas is therefore also less dense around the intruder, and the latter rises due to the buoyancy of the 'bubble' of reduced density it carries around itself. It should be noted however, that there are several mechanisms contributing to the brazil nut effect, like heave processes or convection [47, 48], and a full account of its physics is quite involved.

1.2. *Influence of cohesion*

It is intuitively clear that cohesion between individual grains can alter the physical properties of the granulate dramatically. This has been extensively studied in the past [18, 24, 25, 50–75], with various possible cohesion forces having been taken into account. The most ubiquitous cohesive force one might think of is the van der Waals force acting between adjacent grains. For spherical grains of equal size, the van der Waals binding energy is given by [76]

$$W_{vdW} = \frac{AR}{12s} \tag{3}$$

where A is the Hamaker constant, R is the radius of the grains, and s is their surface separation [77]. The latter will be equal to (or larger than) the amplitude of the surface roughness, δ, which is at least on the order of a micron, usually larger than that. The Hamaker constant is of order 10^{-20} J, for silicon oxide it is approximately 6×10^{-20} J [76]. Even for very small grains with $R = 20\,\mu$m, we readily find that a velocity of one cm/sec (as above) corresponds to a kinetic energy of 5×10^{-16} J, which is orders of magnitude larger than the van der Waals binding energy as given by equation (3). Since the latter scales linearly with the radius, but the kinetic energy scales with its cube, we can conclude that van der Waals energies may be safely neglected for typical grain sizes occurring in granular physics [78].

Clearly, the dominant factors for cohesion are others. Moisture, which is at the focus of the present paper, has a much larger effect. It results in the formation of small capillary bridges between adjacent grains, which exert an attractive force between them by virtue of surface tension. This force is of order $R\gamma$, where γ is the surface tension of the liquid. For grains with a diameter of $100\,\mu$m, it exceeds gravity by two orders of magnitude. As a result, one observes prominent changes, e.g., in the repose angle of granular heaps [65, 79]. In practical systems, like soil, additives may be present which act as a glue [80]. However, in order to deal with well defined systems, we will not consider effects of additives here, but restrict the discussion to ternary systems consisting of the grains, a pure liquid, and the interstitial air.

In the special case where the latter has zero volume fraction, we have a situation as encountered with slurries [81], or sand under water [82]. In the latter case, the behavior is similar to that of dry sand: grains in a homogeneous fluid counter phase. This may even been used as a model system for the dry case [83–86], since no cohesion is present. Although the replacement of air by a liquid comes with a large change in viscosity, its effect is only secondary, and we will disregard this special case here as well. The focus on wet systems has another, serendipitous advantage: the presence of water, which is electrically conductive, saves us the discussion of effects due to electrostatic charges, which may be quite significant, and annoyingly complicated.

2. Capillary forces

When liquid is added to a dry granulate, one observes a transformation from a more or less fluid state to a pasty material, as mentioned in the introduction. We have seen in the previous section that the formation of liquid capillary bridges between adjacent grains certainly plays an important role here, but a satisfactory theory connecting the wetting processes to the mechanical properties of wet granulates has started to emerge only in recent years [18, 24, 25, 61, 70, 71, 74]. One of the simpler questions to be asked is this: is it the (random) network of capillary bridge forces itself which yields the stiffer appearance, or is it just the increase of the friction force between the grains due to their being pushed harder together by the capillary force? More complex questions concern the connectivity of the capillary bridge network, and the transformations of the latter upon shearing the granulate. Once these are sufficiently understood, one may try to understand effects like soil liquefaction, or simply the presence of a yield stress, possibly as dynamic phase transitions in this rather complex non-equilibrium system.

The earliest studies of the influence of liquid capillary bridges on granular materials have been carried out in the context of agglomeration [50, 61, 87]. This is a well established technique used, e.g., in food processing, where fine powders are gradually transformed into granulates with larger particle size. In a standard setup, the powder is 'fluidized' by blowing air from below, and wetted by a liquid dispensed from above as a fine mist through a number of nozzles. As time proceeds, the small powder particles stick together upon collision due to the gluing effect of the liquid, thus forming larger granules.

In order to understand this process, researchers have early started to consider the physics of capillary bridges between grains. A major focus was to predict the tensile strength of the agglomerates which are formed. In model studies, the grains were almost invariably approximated by spheres. Since we are here aiming at a physical picture and are not so much interested in engineering aspects, we will adopt this (at first glance simplistic) view throughout the paper. The discussion of experiments will be restricted to those which have used model systems of spherical particles and may thus be compared with theory. As a phenomenological justification of this approach, one may take the experience that piles of small spheres share most of their physical properties with 'real granulates', like sand, and differ only in some relatively subtle aspects.

2.1. *Capillary bridge between two spheres*

The internal geometry of a pile of grains is characterized by a large volume fraction of empty space (i.e., air) and a large inner surface onto which moisture may condense from the gas phase and form a wetting layer. At the points of contact between adjacent grains, the liquid surface then makes a sharp bend, as shown schematically in figure 2 (top). It is clear that this is not an equilibrium situation: a curved liquid surface gives rise to a Laplace pressure

$$p_L = \gamma(r_1^{-1} + r_2^{-1}) \tag{4}$$

where the r_i are the two principle radii of curvature at the point of interest [76]. The expression $(r_1^{-1} + r_2^{-1})/2$ is called the mean curvature of the surface. Where the liquid films meet, this curvature is very large and negative, such that there is a substantial underpressure which sucks the liquid towards the contact region. Equilibrium is reached when the liquid surface has acquired a spatially constant mean curvature, and forms the equilibrium contact angle, θ, with the surface of the grains [76] as depicted in the center of figure 2. If the grains are subsequently

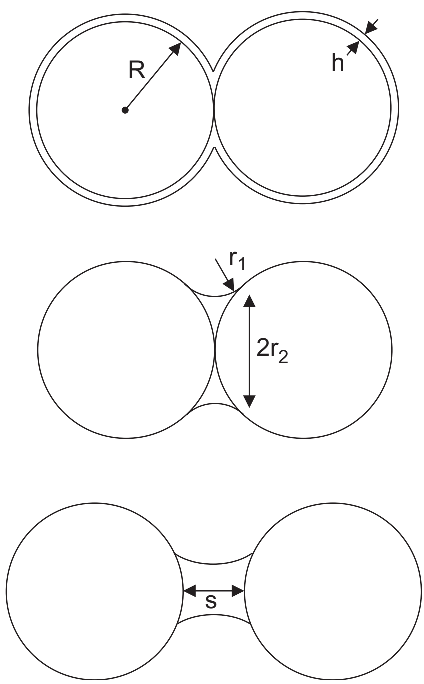

Figure 2. Top: two spherical grains touching, with a wetting layer of thickness h on each of them (thickness exaggerated). Where the liquid film surfaces meet, a sharp bend results which gives rise to a large negative Laplace pressure. Center: equilibrium is reached when a capillary bridge with constant mean curvature has formed (depicted for complete wetting, $\theta = 0$). Bottom: a pendular bridge between two spherical grains. The separation of the grain surfaces is denoted by s. Note that the liquid may make a finite angle θ with the spheres, as in the bottom example, which at equilibrium is Young's contact angle, θ_Y [76].

removed from each other by a distance s, as shown at the bottom of the figure, a so-called pendular bridge results.

If the equilibrium contact angle of the liquid at the grain surface is zero, one speaks of complete wetting. In this case, the surface of the capillary bridge meets the grain surface tangentially, as in the center of figure 2. This is, however, not always the case. In general, the contact angle is given by the Young–Dupré equation [76, 88]

$$\cos\theta = \frac{\gamma_{sg} - \gamma_{sl}}{\gamma} \tag{5}$$

where γ_{sg} and γ_{sl} denote the free energy per unit area of the solid/gas and the solid/liquid interface, respectively. If $\gamma_{sg} < \gamma_{sl}$, there is no gain in free energy upon wetting the grains. Thus the liquid would accumulate in an external reservoir or a drop, rather than invade the granulate. A common example is encountered in the preparation of hot chocolate, when one tries to wet the (hydrophobic) cocoa powder with milk. As one sees from equation (5), the contact angle is larger than 90 degrees if $\gamma_{sg} < \gamma_{sl}$. Trying to form a capillary bridge in a manner analogous to figure 2 (bottom) would necessarily result in a structure with positive Laplace pressure, and there would be no attractive force [89, 90].

It should be noted that gravity can be safely neglected in the discussion of the shape of the bridges. Typical grain sizes are a millimeter or less, such that the size of the bridges is usually in the sub-millimeter regime. This is well below the capillary length l_c of most liquids, which determines the length scale above which gravity plays a significant role in the morphology of liquid structures. It is given by $l_c = \sqrt{\gamma/g\varrho}$, where ϱ is the density of the liquid and g the acceleration due to gravity. For water, $l_c \approx 2.7$ millimeters.

The problem of theoretically determining the shape of the liquid bridge has challenged researchers for a long time. Only in a few special cases does an analytical solution for the liquid surface geometry exist. Most approaches have therefore used approximations. The toroidal approximation used by Fisher [50] gives quite satisfactory results for the attractive force at zero separation s, but fails otherwise [91]. Numerical work [92, 93] made considerable progress later, but left many open questions, in particular concerning the critical distance, s_c, at which the bridge ruptures. A major progress in this respect was a study by Lian *et al.* [94], who reconsidered the problem by means of a variational calculus approach, coupled with numerical methods. They found very good agreement with experimental data obtained by Mason and Clark [95] for the critical rupture distance. Later studies, both theoretical [96, 97] and experimental [90, 98] have corroborated these results.

An excellent and comprehensive presentation of the physics of capillary bridges between spherical bodies has been recently given by Willett *et al.* [90]. This paper provides a large amount of particularly high quality experimental data and demonstrates excellent agreement with state-of-the-art theory. The main facts are as follows. At zero distance, the attractive force exerted by the capillary bridge is given by

$$F_b = 2\pi R\gamma\cos\theta \tag{6}$$

which is equal to the area on the sphere surface covered by the bridge, times the Laplace pressure.

Remarkably, this is independent of the liquid volume, V, of the bridge. It is straightforward to see that this must be so: From the center of figure 2, one sees that $r_1 \propto r_2^2$, or more precisely, $r_1 = r_2^2/2R$ for complete wetting. Since $r_2 \gg r_1$, we have $p_L \approx \gamma/r_1$. The covered area is πr_2^2, such that $F_b = \pi r_2^2 p_L = 2\pi R\gamma$, independent of the size of the bridge. This holds only as long as $r_2 \gg r_1$, so all formulas presented here are valid only if this is fulfilled. However, this is not a serious restriction. We will see below that before the liquid bridge volume becomes large enough to substantially question the validity of equation (6), the bridges coalesce and form more complex geometric structures. The notion of individual capillary bridges then loses its justification.

For the variation of the force at finite separation, exerted by a bridge of liquid volume V,

$$F_b = \frac{2\pi R\gamma\cos\theta}{1 + 1.05\hat{s} + 2.5\hat{s}^2} \tag{7}$$

was shown to be an excellent approximation, with $\hat{s} := s\sqrt{R/V}$. For the critical separation at which the bridge pinches off, the authors find

$$\tilde{s}_c = \left(1 + \frac{\theta}{2}\right)\left(\tilde{V}^{1/3} + 0.1\tilde{V}^{2/3}\right) \tag{8}$$

where $\tilde{s}_c := s_c/R$ and $\tilde{V} := V/R^3$. This is valid for equally sized spheres, with only minute deviations for nonequal ones. In this case, Derjaguin's approximation can be successfully applied, which amounts to averaging the curvatures:

$$\frac{1}{R} := \frac{1}{2}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \tag{9}$$

These formulas give good approximations up to $\tilde{s}_c \approx 0.3$ ($\tilde{V} \approx 0.03$), in the sense that effects which are not considered here, like contact angle hysteresis, give rise to deviations much larger than the errors of equations (7) and (8).

One of the most important features of the inter-particle force due to the capillary bridge is its *hysteretic* nature: for a bridge to form, it is necessary that the grains come into contact, but for the bridge to rupture, a certain finite separation, s_c, must be reached. This will be of major importance further below, when the statistical dynamics and energy flow of the system are discussed.

2.2. *The role of roughness*

It is important to note that equation (5), which defines the theoretical value for the (Young's) contact angle, $\theta = \theta_Y$, is only valid for perfect surfaces. The actual contact angle measured between a liquid surface and its solid support usually deviates from θ_Y due to surface heterogeneities. If the liquid contact line is advancing on the substrate, a larger angle, $\theta_{adv} > \theta_Y$, results, while when it recedes, $\theta_{rec} < \theta_Y$ is observed. The actual contact angle lies somewhere within this interval, $\theta \in [\theta_{rec}, \theta_{adv}]$ (which may span tens of degrees [76]), depending on history. This so-called contact angle hysteresis is certainly relevant for granular matter physics, since the surfaces of grains are always far from perfect. Even if they were perfect at the beginning of an experiment, the unavoidable wear during the inelastic collisions would result in defects and thus eventually in contact angle hysteresis.

Another quite noticeable consequence of the imperfection of the grain surface is that the attractive force, F_b, is not generally independent of the liquid volume, as suggested by equation (6). For very small liquid content, it is intuitively clear that the crevices in the grain surface must first be 'filled' before a bridge can form at all. More precisely, if the roughness amplitude on the surface is δ, the Laplace pressure of the bridge in its early stage of formation is of order $-\gamma/\delta$. If there are dents or crevices with a radius of curvature smaller than δ, these will be filled first, or suck the liquid out of a bridge which is about to form.

This has been elaborated upon in detail by Halsey and Levine a few years ago [99]. They considered the roughness as being characterized by its amplitude, a typical lateral scale, ξ, and a roughness exponent, χ, with $0 < \chi \leq 1$. They identified three regimes, which are denoted by **I**, **II**, and **III**. At very small liquid content (**I**), bridges are formed only between asperities on the grains. The normalized force, $f_b := F_b/2\pi R\gamma$, was then found to vary as

$$f_b(\mathbf{I}) \approx \frac{R^2}{2\pi\delta^2}\left(\frac{R^3}{\delta\xi^2}\right)^{\mu-1}\tilde{V}^{\mu} \tag{10}$$

with $\mu := (2-\chi)/(2+\chi)$. As more liquid is added, a crossover to a new regime (**II**) takes place when $\tilde{V} \approx \xi^2\delta$. In this regime, the statistical roughness of the surface determines the morphology of the liquid structure [99]. It is found

$$f_b(\mathbf{II}) \approx \frac{R^2}{2\pi\delta^2}\tilde{V} \tag{11}$$

If χ is small, as for scratch/dig roughness [100], $\mu \to 1$ and thus $f_b(\mathbf{I}) \to f_b(\mathbf{II})$. If instead $\chi = 1$, we have $\mu = \frac{1}{3}$ and thus $f_b(\mathbf{I}) = (2\pi)^{-1}(\xi/\delta)^{4/3}\tilde{V}^{1/3}$. Finally, when the rough surface is covered with liquid everywhere, we have, in effect, completely wet (i.e., hydrophilic) spheres, which appear smooth and wettable. This leads to the 'ideal' result,

$$f_b(\mathbf{III}) \approx 1 \tag{12}$$

which corresponds to equation (6) for complete wetting. If the contact angle of the liquid with the grain surface is finite, f_b should generally be multiplied by $\cos\theta$.

For practical purposes, it is clear that one would rather like to use an expression for f_b which interpolates between these regimes, in particular since equation (10) through (12) cannot be considered exact anyway, due to the unknown geometry of the roughness. A straightforward answer is

$$f_b \approx \left(1 - e^{-\tilde{V}/\tilde{V}_0}\right)^{\mu} \tag{13}$$

with $\tilde{V}_0$ chosen appropriately. We obtain

$$\tilde{V}_0 = \left(\frac{2\pi\delta^2}{R^2}\right)^{1/\mu}\left(\frac{\delta\xi^2}{R^3}\right)^{(\mu-1)/\mu} \tag{14}$$

which approaches $\tilde{V}_0 = 2\pi\delta^2/R^2$ for small χ. In this practically relevant case, we have the simple result

$$f_b \approx 1 - e^{-V/2\pi R\delta^2} \tag{15}$$

Since the roughness amplitude is usually small compared to the bead radius in the experiments we will discuss here, we see that the transition to regime **III** takes place already at rather minute amounts of liquid. It should again be stressed that the exponential form has been introduced for convenient interpolation only. Other forms, such as

$$f_b = \left(\frac{\tilde{V}}{\tilde{V} + \tilde{V}_0}\right)^{\mu} \tag{16}$$

may alternatively be used, depending on the geometry of the roughness.

If a granulate acquires liquid only through the vapor phase, the adsorbed amounts of liquid are extremely small, corresponding only to molecularly thin films covering the grains. Nevertheless, it is known from everyday experience that humid air can noticeably change the appearance and behavior of a granular material. In this case, regimes **I** and **II** play the dominant role, as extensively studied in recent years [24, 25, 72, 101–103]. The increase of f_b at small V is reflected, e.g., in results obtained earlier by Hornbaker *et al.* who studied the stability and repose angle of granular slopes [65]. Tegzes *et al.* [79, 101] found good agreement with equation (10) through (12) in a similar, but more elaborate study.

However, the details inherently depend upon the geometry of the roughness, which is never precisely known. Furthermore, at this liquid content the granulate has a damp appearance, but has not yet reached what may be called a pasty state, which is suitable, e.g., for sculpturing a sand castle. For studying the latter state, it is most of the time sufficient to consider regime **III** only, with only occasional reference to regimes **I** and **II** where appropriate.

2.3. *Capillary bridge networks*

After sufficient equilibration, the interior of a wet granulate is expected to be characterized by a network of liquid bridges connecting adjacent grains in contact. It is clear that the connectivity of this network of liquid bridges is of importance for the mechanics of the pile [67, 71, 74, 104], be it directly through the capillary force itself, or through the enhancement of the mutual friction forces between the grains due to the increased internal pressure. The connectivity depends upon the packing geometry, which is a complex topic in itself [105–107]. Even for the random close packing of spheres of equal size, which is a comparably 'simple' case, the coordination and nearest-neighbor relations are still under investigation [108, 109].

The first quantitative experimental study of the coordination of randomly packed spheres of equal size has been performed by Bernal and Mason in 1960 [105]. A random pile of steel balls was immersed in black Japan paint. When the paint was drained out of the pile, capillary bridges formed close to the contact points and at points of close proximity (pendular bridges). After the paint had solidified, the pile was taken apart. The points of contact were revealed as annular stains of paint on the balls and could thus be counted. For the random packing which was achieved, with a density (volume fraction of the balls) of $\rho = 0.62$, an average number of $N_c = 6.4$ contacts was obtained for each sphere.

A few years later, Mason analyzed in a similar system the number of neighbors within a certain separation [107]. He found only 5 real contacts per sphere, but about 10 within 0.4 radii of separation, in agreement with earlier work [105]. A theoretical estimate for N_c can be given within the so-called caging model, which constructs a pile of spheres numerically under the condition that each sphere experiences just enough contacts to fix its position in space. A recent study [108] yields $N_c = 4.79$ in this case, which should be considered as a lower bound for real systems. Other reasonings [105, 109] suggest that $N_c \approx 6$. The reported results lie in the range $N_c = 5.6 \pm 0.8$. It should be borne in mind that these studies all considered mono-disperse ensembles, and it is to be questioned whether this applies to granular systems, which always exhibit a certain poly-dispersity.

It is therefore desirable to find a way of looking inside a granular pile, in order to determine the statistics of the network of liquid bridges directly in the system under study. A rather simple way to achieve this goal is to use glass spheres as grains, and to apply an immersion technique to suppress refraction from the glass bead surfaces [111]. Small glass spheres, also called ballotini, can be purchased in large quantities, and with a large variety of sizes and size distributions. Mono-disperse samples (which are quite expensive) are not to be considered as good model systems for granulates. Crystallization may occur, in particular if there are attractive forces between the particles. This will give rise to noticeable side effects which will not be encountered in real systems. We will therefore concentrate on model granulates consisting of glass spheres with a certain poly-dispersity, defined as the relative width of the size distribution. For most (high quality) material commercially available, this ranges from 10 to 20%. Wider distributions can of course be obtained as well. Even for studies not using optical imaging, glass ballotini were used in most cases, such that it appears reasonable here to concentrate on these rather well defined systems.

In a recent study [112], a mixture of toluene and diiodomethane was used as a replacement for the interstitial air. At a volume fraction of 11.9% diiodomethane, its refractive index was 1.51, the same as that of the glass spheres used in the experiment. Perfect index matching cannot be easily achieved, since the refractive index of commercially available glass spheres varies slightly from bead to bead. However, the transparency achieved was sufficient for the purpose of the experiments. Glass spheres were used with mean diameters of 375 μm and 555 μm. As the wetting liquid, small amounts of water were added, with some fluorescein dissolved to ease observation. The aqueous phase wetted the glass beads completely ($\theta \approx 0$) in the presence of the index matching liquid, and formed capillary bridges which could be directly observed and counted.

Samples were prepared by adding the ballotini, and subsequently the stained water, into a glass cuvette containing the immersion liquid. The samples were then shaken for a few minutes until the color due to the added fluorescein, as judged under an ultraviolet lamp, appeared evenly distributed. The volume fraction of the glass beads directly after sample preparation was $\rho \approx 0.57$, as determined by direct observation and counting of beads. This will be referred to as random loose packing below. By gently tapping against the sample, the packing density could be raised to $\rho \approx 0.62$, which is close to the volume fraction of random close packing ($0.64 =: \rho_{rcp}$ [106]) and equal to the density of the mono-disperse steel ball packings investigated by Bernal and Mason [105]. This will henceforth be referred to as random dense packing. It should be mentioned that no detectable dependence of packing densities upon the amount of liquid was observed. This may be attributed to the fact that the capillary forces exerted by the capillary bridges depend only weakly on the bridge volume, as discussed above.

Figure 3 shows optical micrographs of the interior of a sample prepared as described, for two different water contents, W. The latter is defined as the total volume of water divided by the total sample volume (i.e., the total volume occupied by the wet granular pile). The upper image is taken at a water content

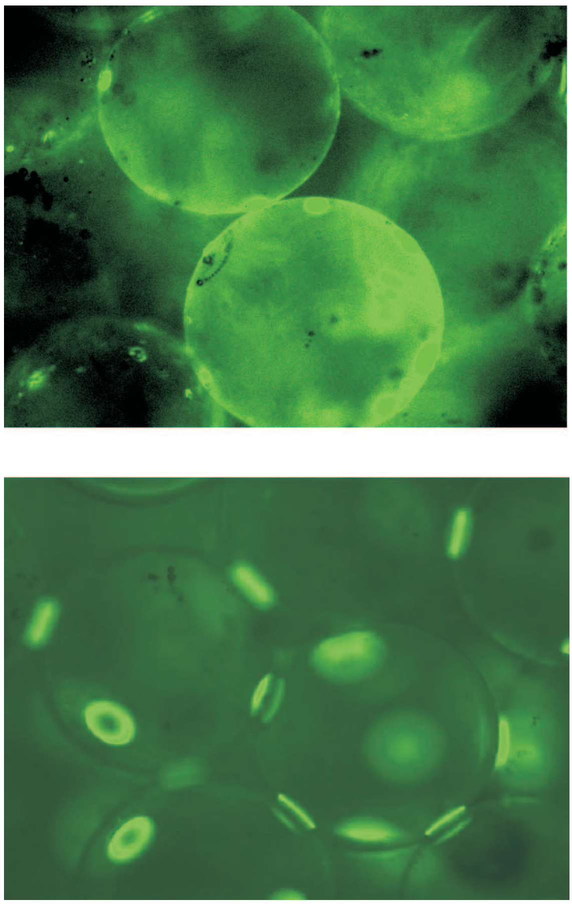

Figure 3. Fluorescence microscope images for different water contents W, with glass beads of average diameter 375 μm. Top: $W = 5.5 \times 10^{-4}$, no bridges have formed yet. All liquid resides within the roughness of the glass surface. Bottom: $W = 2 \times 10^{-3}$, capillary bridges between the beads have formed. The sensitivity of the camera is correspondingly reduced, such that the liquid trapped in the roughness of the glass surface is no more visible.

of $W = 5.5 \times 10^{-4}$, before bridges are seen. The stained liquid on the surfaces of the spheres is clearly visible. In the lower image, at $W = 2 \times 10^{-3}$, bridges have clearly formed. The sensitivity of the camera had to be reduced to avoid over-exposure. The film on the surface, which is still present, is therefore not visible in this image. The threshold for bridge formation was determined to be $W_0 \approx 7 \times 10^{-4}$.

We shall compare this to what we would expect on the basis of the roughness of the beads, which can be determined by scanning force microscopy. In this case, a roughness amplitude of $\delta \approx 500\,\text{nm}$ was measured. The maximum amount of water which can be stored per sphere in this roughness is about $4\pi R^2 \delta$. This corresponds to a total water content of $W_{max} = 3\rho\delta/R$ [113], which yields 5×10^{-3}. This is in fact larger than the observed onset of bridge formation, which is expected since the bridges may be in successful competition for liquid with the roughness well before W_{max} is reached.

Once the bridges are fully developed, their average number per sphere, N, can be determined precisely by zooming with the microscope through the sample. This was carried out at variable liquid content for random loose packing ($\rho = 0.57$) as well as for random dense packing ($\rho = 0.62$). It is particularly instructive to consider N as a function of the pinch-off distance, $\tilde{s}_c$ (cf. equation (8)), as calculated from the liquid content. As it turns out [112], it can be quite well represented by

$$N \approx N_0(\rho) + 6\,\tilde{s}_c \tag{17}$$

N_0 was found to be 5.1 for $\rho = 0.57$ and 5.9 for $\rho = 0.62$. It is self-evident to identify N_0 with N_c discussed above. The corresponding slope $dN_0/d\rho$ is ≈ 16, which compares well with what is known for the number of real contacts in random sphere packings of variable density [114]. The coefficient 6 of the second term corresponds to the number of pendular bridges, which is expected to be about half the number of contacts with separation $\tilde{s} \leq \tilde{s}_c$ [112]. In fact, Mason [107] found a slope $dN/d\tilde{s}$ in this region of about 12. A similar result is obtained with the caging model [108].

Let us finally compare the liquid content at which regime **III** is reached to the maximum liquid content which can be stored in a granular pile. The latter is a significant fraction of the sample volume, at least 36 % (for $\rho = \rho_{rcp}$). Since regime **III** is reached already at a liquid content well below one percent (for spherical grains), it is obvious that this regime **III**, which corresponds to the pasty state, is very relevant for this class of materials. As the above considerations show, its internal geometry may be well described using general results obtained for random packings of equally sized spheres. One may hope that for pronouncedly non-spherical grains, the theory of packing of analogous model systems [109] may be applied, but this remains to be investigated.

Once we know how many bridges are in contact with each sphere, we can estimate the average capillary bridge volume as a function of the liquid content, W. We have $N \approx 6(1 + \tilde{s}_c)$ and

$$\tilde{V} = \frac{8\pi W}{3\rho N} \tag{18}$$

if we assume all liquid to go into the capillary bridges. In this case, using $\tilde{V}_0 = 2\pi(\delta/R)^2$, we would expect $W_0 \approx 3 \times 10^{-5}$, which is certainly a lower bound,

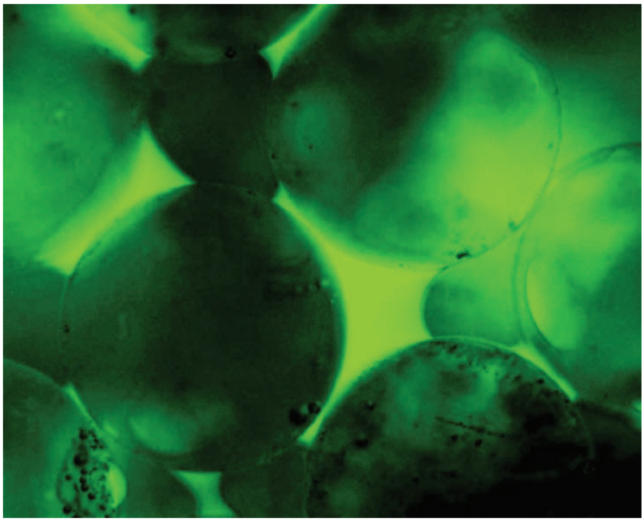

Figure 4. Liquid aggregates forming at higher liquid content ($W = 0.03$).

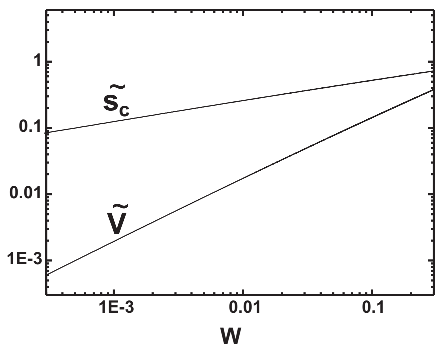

Figure 5. The critical separation and the bridge volume, both normalized, as a function of the liquid content. For this graph, it is assumed that all liquid is stored in the capillary bridges. The solid lines are thus to be considered as upper bounds.

and indeed well below the experimental value. W_0 should thus be considered as an experimentally determined parameter.

Together with equation (8), equation (18) can be numerically solved. The result is depicted in figure 5. We see that the range of validity of equation (6) to (8) is left at a liquid content of about 2 percent, when $\tilde{s}_c$ becomes significantly larger than 0.3. Furthermore, liquid bridges occasionally coalesce to liquid aggregates in this range. An example is shown in figure 4 for $W = 0.03$. We will henceforth call this regime, where capillary bridges coalesce to larger aggregates, regime **IV**. Since the coalescence of bridges will finally be accompanied by a reduction of liquid surface, we might expect a reduction of the yield stress of the granular pile in this regime.

Finally, the liquid aggregates will undergo a percolation transition [115]. When this has happened, a *paramaecium* cell would be able to swim from one end

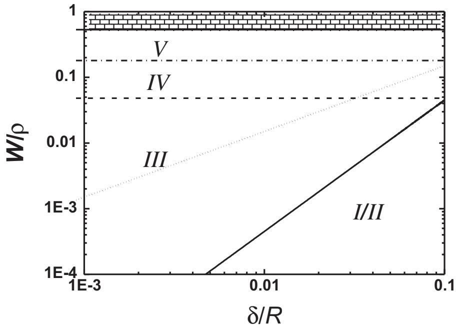

Figure 6. The various regimes in the plane of normalized liquid content, W/ρ, and normalized roughness amplitude, $\tilde{\delta} = \delta/R$. The solid line indicates the crossover from regimes **I** and **II** to regime **III** at static conditions. Dynamics may shift this up to the thin dotted line, which represents the initial bridges only (V_1 in equation (19)). Regimes **IV** and **V** correspond to coalesced bridges and to percolated liquid clusters, respectively. All lines are calculated for $\rho = \rho_{rcp} = 0.64$. The shaded region corresponds to the (unphysical) case $W > 1 - \rho$ (more liquid than interstitial volume).

of the sample to the other without leaving the water. This is regime **V**. In this case, liquid transport over large distances can be induced by minute variations in the packing geometry in different parts of the sample. The physical description of this state appears quite cumbersome [116]. It should be noted in passing that in particle technology and other areas of engineering, a different terminology is commonly used. There one distinguishes between pendular, funicular, and capillary states of a wet granulate [87, 117]. At first glance, 'pendular' and 'funicular' seem to correspond roughly to regime **III** and **IV**, respectively. However, the transition between pendular and funicular is sometimes reported around 10% of liquid content, which is already within regime **V** in our terminology. In this paper, we will use the latter only.

Figure 6 shows all of the regimes discussed above in the plane spanned by the normalized liquid content, W/ρ and the normalized roughness amplitude, $\tilde{\delta} = \delta/R$. The solid line indicates the crossover from the roughness dominated regime (**I** and **II**) to the 'ideal' regime (**III**), where equations (6) to (8) hold. It corresponds to a crossover taking place at $\tilde{V}_0 = 2\pi\tilde{\delta}^2$ as described in the previous section. The dashed and dash-dotted lines mark the (approximate) transitions between the various regimes corresponding to the 'pasty state'. The large region denoted **III** is certainly the most well defined among the regimes included in the diagram. The thin dotted line indicates dynamic effects, which are to be discussed in the next section.

2.4. *Dynamics*

So far, we have only been concerned with static aspects of capillary bridges. However, for a thorough understanding of what we have above called the pasty state of granular matter, dynamic properties will be of great importance. We thus have to consider as well the dynamic aspects of the individual liquid bridges. There are two main issues to address. One is the dynamics of bridge formation which influences the amount of liquid effectively participating in the impact kinetics of the grains. The other is the dissipation within the liquid itself, i.e., viscous forces.

For a capillary bridge to form, two grains must come into contact. More precisely, the surfaces of the liquid adsorbed on them must touch each other. When this happens, a small initial bridge forms more or less instantaneously from the liquid in the contact region. Its volume, V_1, may be obtained from the average liquid surface coverage, h, which is related to the total liquid content by $h = WR/3\rho$ (cf. figure 2). In regime **III**, h is of the same order of magnitude as the roughness amplitude, δ. The contact region comprises an area of order $2\pi Rh$, such that the initial bridge volume is $V_1 \approx 8\pi Rh^2$ [118]. For the pinch-off distance, we thus obtain (cf. equation (8)) $s_c \approx V^{1/3} = h(8\pi R/h)^{1/3}$. Since $8\pi R \gg h$, we clearly see that even for this initial bridge, the critical distance at which rupture occurs is significantly larger than the adsorbed film thickness, which sets the scale for the proximity necessary for bridge formation. This means that the hysteretic nature of the interaction force is already fully developed.

Once this initial bridge has been established, liquid starts to accumulate near the point of contact by virtue of the gradient in Laplace pressure (equation (4)). This can proceed either through the adsorbed liquid film [119, 120], or through the gas phase [121, 122] by evaporation and recondensation. It is difficult to assess the relative contributions of these two mechanisms in the real system. In particular, the roughness may influence the transport within the film noticeably [123–126]. It is therefore indispensable to study this process empirically. This has been done recently by Kohonen *et al.* [113], who investigated the growth of capillary bridges between glass beads by optical microscopy with a number of different liquids. It was found that the temporal evolution of the bridge volume followed, within experimental scattering, an exponential:

$$V(t) = V_1 + V_2(1 - e^{-t/t_0}) \tag{19}$$

where V_2 and t_0 are constants. t_0 scales with the viscosity of the liquid, η, and is about five minutes for water at room temperature. As a consequence, the boundary to regime **III** may be shifted in the dynamic case ($t \ll t_0$) up to the thin dotted line in figure 6.

For several simple liquids, it has been shown that the equilibration dynamics through liquid wetting layers on surfaces with some roughness is remarkably weakly dependent on the adsorbed film thickness [125, 126]. It was found that effects of roughness and of dispersion forces partly cancel each other, such that the decay time of lateral variations in film thickness is quite constant over a wide range of adsorbed liquid coverage. For simple liquids like ethanol and propane, the lateral transport coefficient a, which scales as η^{-1}, was found to be of order 10^{-10} to 10^{-11} m^2/sec. The corresponding time scale for liquid transport is $2\pi a/R^2$, which then is of order 10^{-2} to 10^{-3} sec^{-1}. This is remarkably close to t_0^{-1} as reported above. It is thus natural to propose that t_0 should scale as ηR^2, but be only weakly dependent on the amount of adsorbed liquid. This might be tested in forthcoming studies.

It should be noted that there may be still a further increase, at times large as compared to t_0, which leads to long-term aging effects not included in equation (19). An upper bound for V_2 can obtained via $V_1 + V_2 \leq V(\infty) = 8\pi R^2 h/N$, where N is the number of bridges per sphere as above. This yields $V_2 \leq V_1(R/Nh - 1) \approx V_1 R/Nh \gg V_1$. V_2 can obviously be much larger than V_1. Aging effects are an interesting topic in themselves, in particular with respect to applications, and have been eagerly studied [24, 25, 72, 102, 103].

In any case, we see that there is considerable redistribution of liquid after the contact has been established between adjacent grains. Since this affects the bridge volume, it also affects the critical distance s_c at which the bridge pinches off. However, s_c increases only as the cube root of V. According to the above expressions, it therefore changes, as time proceeds, by not more than about a factor of three. It is furthermore instructive to consider how this affects the energy required to rupture the bridge. This is obtained by integrating the r.h.s. of equation (7), which yields

$$\Delta E_{cap} = \int_0^{s_c} F_b ds \approx 5.5\sqrt{\tilde{V}}\ \gamma R^2 \cos\theta \tag{20}$$

This scales approximately as $\tilde{s}_c^{3/2}$, and therefore changes by up to a factor of five during the growth of the bridge from V_1 to $V(\infty)$.

The possible influence of the viscosity of the liquid has been studied a lot, in particular with respect to agglomeration processes [61, 66, 73, 127]. Ennis *et al.* [61] have discussed this in terms of the capillary number, $Ca := v\eta/\gamma$, where v is the relative velocity of the grains at impact. They found that the viscous force is given by

$$F_{visc} \approx 2\pi R\gamma \frac{3Ca}{4\tilde{s}} \tag{21}$$

which diverges as $\tilde{s} \to 0$. However, this divergence is not of practical relevance, since the roughness amplitude can be considered as a lower bound for s. Assuming the velocity to be roughly constant, the total energy loss due to viscosity is obtained as

$$\Delta E_{visc} = \int_\delta^{s_c} F_{visc} ds \approx \frac{3\pi}{2} R^2 \eta v \ln\frac{\tilde{s}_c}{\tilde{\delta}} \tag{22}$$

This is to be compared to the energy loss due to the capillary force, ΔE_{cap} (equation (20)). The ratio of the two loss terms is [128]

$$\frac{\Delta E_{visc}}{\Delta E_{cap}} \approx \frac{0.86Ca}{\cos\theta}\ \tilde{s}_c^{-3/2} \ln\frac{\tilde{s}_c}{\tilde{\delta}} \tag{23}$$

We see that the influence of viscous forces increases linearly with the capillary number, and thus with the impact velocity. As the latter is gradually increased, a transition from a regime dominated entirely by capillary force to a viscous regime will take place. The crossover may be defined by $\Delta E_{visc} = \Delta E_{cap}$, and is indicated as the solid curves in figure 7 for different values of $\tilde{\delta}$. The latter increases from 0.01 to 0.05, shifting the curve from bottom to top.

It is instructive to put in a few numbers as encountered in real systems. The capillary velocity, $v_{cap} := \gamma/\eta$, for water at room temperature is 70 m/sec. Typical relative velocities in fluidization experiments are cm/sec, those in creep and similar processes are much below that. We thus can conclude that for aqueous systems, $Ca = v/v_{cap}$ is small enough for viscosity to be neglected. On the contrary, v_{cap} for glycerol is only 4 cm/sec, which is of the same order as experimental velocities, and viscosity must be taken into account [129].

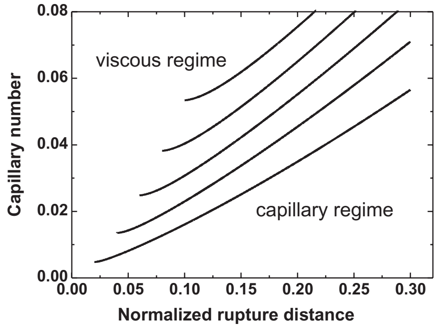

Figure 7. The boundary between the capillary and the viscous regime in the $Ca - \tilde{s}_c$-plane, for various values of the roughness amplitude. From bottom to top, the curves correspond to $\tilde{\delta} = 0.01$, 0.02, 0.03, 0.04, and 0.05. Complete wetting is assumed ($\theta = 0$).

3. Theoretical modelling

We are now in shape to discuss how the main ingredients of the microscopic aspects of wet granulates may be cast into a theoretical model which allows for studies in the spirit of statistical mechanics. The goal must be to make the model as simple as possible, but as complex as necessary to exhibit the main features observed with wet granular matter in experiments. Many attempts have been made in the past. The interaction between neighboring grains has been modelled as 'spring and dashpot' in the simplest cases, but elaborate models have also been proposed. The important role of the hysteretic nature of the capillary force seems to have been appreciated for the first time by Simons *et al.* [96], and was later elaborated upon in a detailed simulation of wet granular fluidization via gas flow [67]. In response to the recent progress concerning regime **III**, we will concentrate here on the role of hysteresis as the most important feature governing the physics of these systems. As we will see, it leads to a dynamical phase transition from a quiescent to a fluid state, quite reminiscent to yield stress effects, or liquefaction phenomena in soil mechanics.

3.1. *Wet granular matter in phase space*

In *dry* granular physics, the restitution coefficient, $\epsilon < 1$ is usually taken as a constant number characterizing the granulate under investigation. This will henceforth be referred to as the *standard model* of granular physics. In more refined models, the restitution coefficient becomes a function of the initial kinetic energy, E_i, of the collision partners. One finds for the rather general case of viscoelastic particles that ϵ is a monotonously *decreasing* function of E_i. It can be described as [32]

$$\epsilon = 1 - C_{ve} E_i^{1/10} \tag{24}$$

in reasonable approximation, where C_{ve} is a constant. This will be called the *viscoelastic model* below.

A qualitatively different situation arises with hysteretic liquid bridges, which will be referred to as the *capillary model* in what follows. In this case, the energy lost in

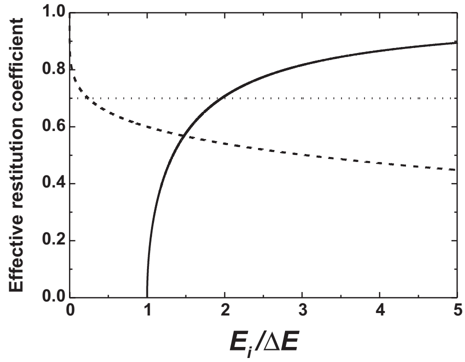

Figure 8. Effective restitution coefficients vs. impact kinetic energy for the standard model (dotted), the viscoelastic model (dashed) and the capillary model (solid). The latter exhibits a zero at finite energy.

a collision is $\Delta E = \Delta E_{cap}$ (cf. equation (20)), which is roughly constant. Since $\Delta E = (1 - \epsilon^2)E_i$, we have

$$\epsilon = \sqrt{1 - \frac{\Delta E_{cap}}{E_i}} \tag{25}$$

This is a strongly *increasing* function of E_i, as opposed to the behavior encountered in the viscoelastic model. Furthermore, with larger velocity there is less time for bridge formation, and ΔE_{cap} becomes smaller. This gives rise to an even steeper increase of ϵ with increasing energy. In figure 8, the effective restitution coefficients are depicted schematically as a function of the impact kinetic energy, E_i, for the standard model (dotted line), the viscoelastic model (dashed curve) and the capillary model (solid curve, equation (25)). The latter exhibits a zero at $E_i = \Delta E$. For $E_i < \Delta E$, the particles remain glued together by the liquid after the impact. In principle, there is a zero also for the viscoelastic model, but this occurs at enormously high energy ($\approx 10^4$ in our example), where the validity of the model may be questionable. In the range of physically reasonable energies, the zero of ϵ can be considered a particularity of the capillary model.

It should be noted in passing that systems of grains with a possibility of sticking together are also attracting considerable interest in the context of structure formation in the universe [130–134]. They are referred to as 'sticky dust' or 'sticky gas' in this context, and play a significant role in the formation of planetesimals and planets. This demonstrates the ubiquity of the systems we are discussing here, and the significance of investigating their properties from a statistical mechanics point of view.

With the separation s of the two impacting grains as a spatial coordinate, we can easily construct a qualitative phase portrait of the impact process, as shown at the top of figure 9. Since the grains are considered here to be ideal spheres, rotational degrees of freedom are disregarded. The motion is completely free of force, except a single point on the time axis, when the impact takes place. In the viscoelastic model,

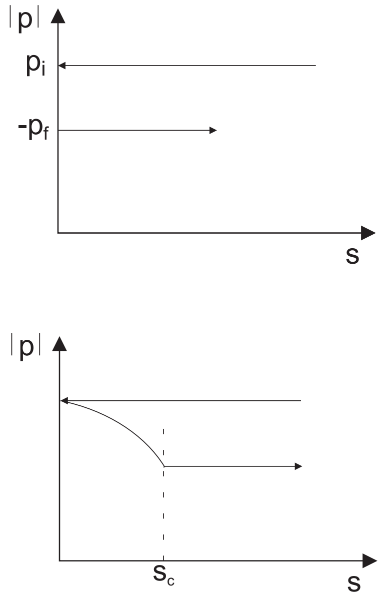

Figure 9. Schematic phase portrait of an impact of two grains in the standard model (top) and in the capillary model (bottom). Note that on the vertical axis the absolute of the momentum is plotted, which is positive before and after impact.

this becomes a short but finite time interval, during which the trajectory rushes from $(0, p_i)$ to $(0, -p_f)$, but effectively, the picture is unchanged. In the capillary model, however, the process of energy dissipation is extended over the full interval $s \in [0, s_c]$, as shown at the bottom of the figure.

The system may be described by a history-dependent Hamiltonian, which is

$$H = \frac{p^2}{2m} + M(t)\int_0^s F_b(s')\, ds' \tag{26}$$

where $M: \mathrm{R} \to \{0, 1\}$ is just switching the capillary term on and off. In particular, $M = 1$ immediately after the contact of the spheres, but $M = 0$ for spheres approaching from remote. If we were to set $M = 1$ whenever $s \leq s_c$, as has been done before [66], we could write M as a function of s, hence a Hamiltonian system in the usual sense would result. In this case, there would be no dissipation at all in our model system. Note that the collisions themselves are considered elastic, such that for $s_c \to 0$ we do not obtain the standard model, but the hard sphere gas. If we used a random function for $M(t)$, there would probably be no dissipation either, on average. The dissipative character of the system comes about by the correlation of $M(t)$ with the course of the system in phase space.

Since it is obvious that the mere presence of hysteresis is the main feature to investigate here, we make another step of simplification and introduce what we call the *minimal capillary model*, assuming that the force is independent of the separation, s, until the bridge ruptures. We have shown above that this introduces a change in the numerical value of ΔE, but on average is not expected to give rise to qualitative changes in the behavior of the system. We then have

$$H = \frac{p^2}{2m} + M(t)F_b s \tag{27}$$

where F_b is the (constant) bridge force. The energy which is lost whenever a bridge is formed, elongated to $s = s_c$, and ruptured is now simply given by $\Delta E = F_b s_c$.

It is instructive to compare the standard model and the minimal capillary model in terms of the phase portraits close to an impact. In the standard model, the trajectories are contracted in the momentum direction by the factor ϵ, as obvious from figure 10. If the distance in momentum direction before the collision is δ_i, we have after the collision $\delta_f = \epsilon\delta_i$. Due to the momentum reduction, the velocity in phase space is also reduced by ϵ, since $ds/dt = p/m$. We see that the phase flow is contracting in both the longitudinal and in the transverse direction.

The phase portrait of two neighboring trajectories in the minimal capillary model is depicted in figure 11. It is elementary to show that

$$\frac{\delta_f}{\delta_i} = \frac{1}{\sqrt{1-(\Delta E/E_i)}} \tag{28}$$

which is independent of the form of $F_b(s)$, and therefore also valid for the full model (equation (20)). Obviously, the phase flow is transversely expanding, in strong contrast to the standard model.

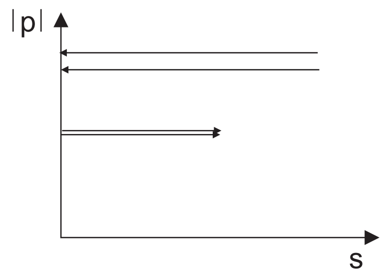

Figure 10. Schematic phase portrait of an impact of two grains in the standard model, with two neighboring trajectories.

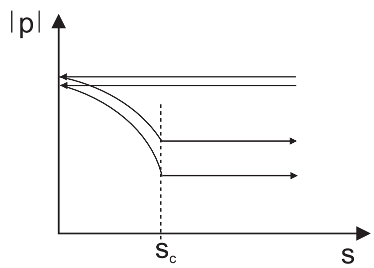

Figure 11. Schematic phase portrait of an impact of two grains in the minimal capillary model, with two neighboring trajectories.

A more formal presentation of the transverse spread of the phase flow uses the Lyapunov spectrum of the system. The Lyapunov exponents, λ, describe the temporal evolution of the distance $d(t)$ of two closely neighboring points in phase space. Since for sufficiently small distances the dynamics can be regarded as linear, the distance is exponential in time, $d(t_1) = d(t_0)exp[\lambda(t_1 - t_0)]$. We thus can write

$$\lambda \sim \frac{\ln d(t_1) - \ln d(t_0)}{t_1 - t_0} \tag{29}$$

and bear in mind that λ is to be defined as a time average. It is clear that in $6n$-dimensional phase space, there are $6n$ such exponents, and these are the eigenvalues of the linear map of the distance vector, $\mathbf{d}(t)$, provided by the phase flow along the trajectory.

In both models, we will have the spatial contributions to the λ_i as known from the hard sphere gas [135, 136] (which are positive), plus in addition the effect of the dissipative impact kinetics. In the standard model, the time-averaged contribution of the impact kinetics to the Lyapunov exponents can be obtained by taking d_i and d_f for $d(t_0)$ and $d(t_1)$, respectively, and replacing the denominator in equation (29) by the inverse impact frequency:

$$\langle \lambda_i \rangle \sim \frac{\nu}{6n} \ln \epsilon \tag{30}$$

Here we have divided by $6n$ because each λ_i concerns only one of the (p, s)-hyperplanes in phase space. ν is the impact frequency, which can be estimated from the mean particle velocity, $\langle v \rangle$, by $\nu \approx n\langle v \rangle / l$, where l is the mean free path. Since $\epsilon < 1$, this contribution is always negative. The sum of all these contributions, $\Lambda := \sum_i \lambda_i$, is also of interest. We obtain $\langle \Lambda \rangle \sim \nu \ln \epsilon$ which is negative as well. It should be mentioned here again that there is a 'background' of Lyapunov exponents from the hard sphere dynamics. It is a difficult task to decide whether Λ will be positive or negative if all processes are taken into account [136]. What we can say, however, is that the collisions, where the dissipation really takes place, give only negative contributions to the λ_i in the standard model.

Now let us consider the minimal capillary model. From equation (28), we can derive the average transverse contribution to the Lyapunov exponents as

$$\langle \lambda_T \rangle = -\frac{\nu}{6n} \ln\left(1 - \frac{\Delta E}{E_i}\right) \tag{31}$$

which are obviously positive [137, 138]. The divergence, which also appears in equation (28), will be smeared out due to the statistical distribution of impact energies.

It is interesting to consider this result in the context of the so-called fluctuation theorem [139–142], which has attracted substantial interest recently. It is a general statement on the statistics of fluctuations in non-equilibrium systems. It has been derived from phase space considerations and states that

$$\frac{p(-j, \tau)}{p(j, \tau)} = \exp(-nd\alpha\tau) \tag{32}$$

where $p(j,\tau)$ is the probability to observe an average entropy current j when averaging over the time τ, d is the spatial dimension, and α is the sum over all Lyapunov exponents. Positive entropy currents are those which follow the applied field. In the Rayleigh–Benard–Cell, for instance, j is positive if the heat flows from bottom to top. $p(-j,\tau)$ then gives the probability of a violation of the second law of thermodynamics during a limited time τ.

In fact, it is quite easy to see how a form

$$\frac{p(-j,\tau)}{p(j,\tau)} = \exp(-n\beta\tau) \tag{33}$$

may arise. The entropy current should fluctuate around a positive mean value, $\langle j \rangle$. If the distribution of $p(j)$ is Gaussian, equation (33) is immediately obtained. However, it is not at all trivial that this also holds for non-Gaussian distributions, leave alone that $\beta = d\alpha$. Since α involves the Lyapunov spectrum of the system, it is particularly tempting to apply this theorem to wet granular systems described by the capillary model. A few studies have recently applied the fluctuation theorem to dry granular systems [143, 144], but, to the best of the author's knowledge, there was no such investigation yet for a wet system.

3.2. *Wet granular matter in silico*

The most striking feature of the interaction by capillary forces, at least in regime **III**, is its hysteretic nature. Since there is no qualitative difference in the behavior of wet sand and wet glass beads, one is tempted to assume that the rougher shape of natural grains is not essential for the main properties of the system, although it surely changes some characteristics quantitatively. Thus it may be taken as a hypothesis that the hysteretic nature of the interaction is sufficient for explaining the main mechanical features, both static and dynamic, of a wet granulate in capillary regime **III**.

In order to test this hypothesis, one must prepare a granulate which has no other interactions than the hysteretic force. This cannot be done experimentally (because all real beads exhibit friction), but it can be done on a computer. In a recent study [145], a system consisting of several hundred spheres was investigated, which experienced only nearest neighbor central forces, i.e., there was no friction between adjacent grain surfaces. The spheres chosen were slightly poly-disperse, as in typical experiments, in order to avoid spurious crystallization effects. The density was chosen close to random dense packing. The system was placed in a cubic box with periodic boundary conditions in all three dimensions. An attractive force, F_b, was introduced as for the minimal capillary model discussed above, with bridges forming instantaneously upon contact.

A shear field was applied to the whole system, resembling a gravitational field with oscillatory spatial variation. This force field was $\vec{e}_z F \cos(2\pi x/L)$, where L is the lateral dimension of the box. There was thus a shear field within the granulate, while the total force on the system was (very close to) zero. Simulations were run at different values of F, and the motions of the spheres were recorded. It proved particularly instructive to plot the mean square displacement of the spheres as a function of simulation time. As it is shown in figure 12, roughly straight lines

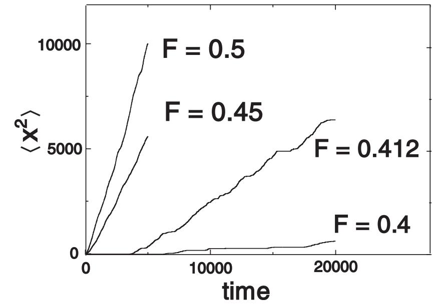

Figure 12. The mean square displacement of the spheres in our simulation of a granular pile under external shear stress for different values of F, given in units of F_b. For larger values of F, a linear behavior is found, indicative of diffusive transport.

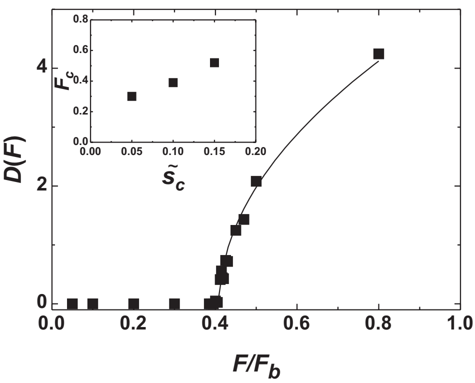

Figure 13. The ensemble averaged diffusion coefficient of a simulated pile of spheres. The solid line represents the form $D = D_0(F - F_c)^{1/2}$.

were found for values of F well above 0.4. This points to a diffusive behavior. At least in this regime, the motion can thus be characterized by a diffusion constant, $D := d\langle x^2\rangle/dt$. Below $F = 0.4\ F_b$, only small scale motion was present. As long as there are no topological transformations in the neighborhood relations, no bridges are broken, and the system is Hamiltonian. Remember that the liquid is assumed inviscid in the simulation, such that there is no dissipation other than due to bridge rupture. Thus there is a small scale Brownian-like motion which persists throughout the simulation time by energy conservation.

When the diffusion coefficient D is plotted as a function of the applied shear force F, critical behavior becomes obvious. As figure 13 shows, there is almost no diffusive motion below a critical force of about $F_c \approx 0.41\ F_b$. Close to this critical force, strong avalanching effects within the system were observed. The average energy of these avalanches diverged algebraically at $F = F_c$ [145], but diffusive transport remained slow. Above F_c, the latter builds up sharply, as can be seen from the figure. This behavior is well resemblant of phenomena like soil liquefaction or land slides, where a wet granulate becomes fluid abruptly under the influence of external shear.

For $F \geq F_c$, our data can be well fitted with a critical exponent of 0.5, as indicated by the solid curve.

Let us briefly outline a microscopic picture which may explain this result. The distance a sphere traverses during a site exchange process scales with the sphere radius, R. The energy the sphere takes up from the applied shear field during that process is thus of order R^2F/L. At the same time, a certain number, G, of capillary bridges must be broken. G is a geometrical factor of order unity which depends upon the packing density. The total energy which must be afforded to break these bridges is Gs_cF_b. The kinetic energy which is left is afterwards randomized into disordered motion by subsequent impacts. The granular temperature thus corresponds to $E_{kin} := (R^2F/L - Gs_cF_b)$. The diffusion constant is equal to the average jumping distance, times the particle velocity. The latter is given by $\sqrt{2E_{kin}/m}$. If we identify this with the randomized energy above, we have immediately

$$D = D_0(F - F_c)^{1/2} \tag{34}$$

with $F_c = Gs_cF_bL/R^2$. The critical exponent of 0.5 is indeed nicely consistent with the simulation results.

The critical force is predicted above to scale proportional to the rupture distance, s_c. In the inset of figure 13, it can be seen that this is at least qualitatively the case [112]. In fact, it is expected to scale sub-linearly (as observed), since the number of bridges broken upon site exchange will decrease with increasing s_c. In the simulations, s_cL/R^2 was of order one, such that $G \approx 2$ would explain the data, which appears quite reasonable. Finally, it should be mentioned that the critical force turned out to be very sensitive to the packing density. It increased strongly as ρ came closer to ρ_{rcp}, and seemed to diverge logarithmically at $\rho_c \approx 0.65$ [146]. These simulations thus corroborate the idea that the simple picture put forward above as the minimal capillary model may well account for at least some of the main features observed with wet granulates in regime **III**.

4. Experimental studies

4.1. *Avalanching and repose angle*

Returning to the hourglass with which we began, one observes that in the lower of the two glass containers a conical heap of sand forms with a rather well defined cone angle. As it continues to accumulate sand on its tip, the cone angle exhibits a steady increase until it reaches a maximum angle of stability, Θ_m, and then decreases abruptly in an avalanche which takes it back to the so-called repose angle, Θ_r. This continues as long as sand is added, with the actual angle oscillating between Θ_m and Θ_r in the manner just described. It is easy to imagine that both Θ_r and Θ_m are very sensitive to cohesion between the grains, i.e., the presence of liquid, or other parameters. In fact, by proper choice of material and conditions, one may achieve $\Theta_m = \Theta_r$, or make the difference very large, up to tens of degrees. Systematic studies of the angle of repose as a function of humidity have in fact been carried out by various groups in the recent past [24, 25, 65, 101]. These studies used glass spheres throughout, and found that the results were not qualitatively different from non-spherical granulates like regular sand. Hornbaker *et al.* [65] as well as

Tegzes *et al.* [101] found a linear increase of Θ_r as a function of humidity. This regime was shown to be related to the roughness of the grain surface [99, 101]. Measurements of the repose angle and the maximum angle of stability have also been very useful to study aging of granular piles in humid atmosphere [24, 102, 103].

Most of these studies have been performed at extremely small liquid content. Those who extend into regime **III** [79, 101, 147] show nice agreement with the predictions listed above on the basis of the capillary bridge forces. The validity of equations (10) to (12) was demonstrated in a unprecedentedly wide range of W [101]. Detailed analyses of Θ_r and Θ_m have been carried out in so-called rotating drum experiments, in which the granulate occupies part of a hollow cylinder which slowly rotates around its (horizontal) axis [79, 147]. The angle of the pile surface then oscillates between Θ_m and Θ_r as in the hourglass. One of the results was that for a wet granulate, Θ_m increases significantly as the speed of rotation is decreased. This is qualitatively what is expected on the basis of the gradual increase of the liquid bridge volume after the formation of a contact [79], cf. equation (19).

4.2. *Experiments with defined shear*

One of the most straightforward questions to pose concerns the yield stress of the material: what shear stress is necessary to sustain a certain shear rate? The main experimental problem here is the occurrence of avalanches, the formation of fault zones, and arching, which make it difficult to establish a spatially homogeneous shear rate extending over the whole sample. In experiments concerning the critical angle of granular piles, Θ_m, one is certainly far from a homogeneous shear field, but the above mentioned problems also play a role in most of the experiments which apply the shear by moving rigid boundaries [15, 18, 19].

In order to avoid such avalanches, triaxial shear cells have been routinely used in the past [4]. Recently, a different scheme has been used to apply the shear in a rather homogeneous way [112, 113]. Figure 14 shows a schematic of the experimental setup. The granular material (spherical glass beads as above) was filled into a cylindrical cell with a diameter of 2 cm and a length of 1 cm. The flat sides of the cell each consisted of a thin latex membrane. Adjacent to each membrane the cylinder continued as a water reservoir which was connected to a syringe. The pistons of the two syringes could be moved by computer-controlled stepper motors. When the two

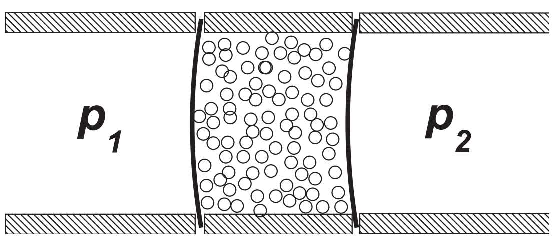

Figure 14. Sketch of the experimental setup. The thick solid lines indicate the latex membranes, which separate the sample from the adjacent water reservoirs. The granulate is sheared with a mean velocity u by changing the content of the two reservoirs. The pressures p_1 and p_2 are continuously recorded.

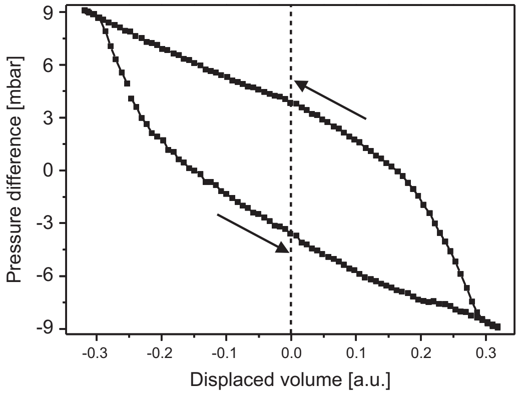

Figure 15. A typical hysteresis loop as obtained with the setup sketched in figure 14.

pistons were moved at equal speed but in opposite direction, the sample between the membranes was deformed at constant volume and well defined speed, u. The latter is defined as the rate of change of the reservoir volume, divided by the cross section of the cylinder. The elastic tension of the membranes ensured a (roughly) parabolic deformation, and thus a flow profile similar to Poiseuille flow within the sample. In this way, avalanching and arching can be kept to a tolerable minimum, and a comparably homogeneous shear deformation of the sample as a whole was achieved.

The pressures in both reservoirs, p_1 and p_2, were recorded while the sample was sheared back and forth with a constant velocity u. A plot of the pressure difference, $\Delta p := p_1 - p_2$ as a function of the displaced volume (or strain) yielded a well developed hysteresis loop, as shown in figure 15. The slope of the two branches of this loop corresponds to the elastic response of the membranes. The opening of the loop, $\Delta p(0)$ (taken at the dotted line), is a measure of the yield stress at the applied shear rate. The latter was small (below 1 millimeter per second), such that the sample was always well within the capillary regime (cf. figure 7). The total strain was chosen large enough to induce irreversible particle motions [148].

This setup allows a number of important questions to be addressed. First of all, as mentioned above, it is not clear by which mechanism the capillary bridges change the mechanical properties of the granulate to the pasty appearance observed. If it was by increasing the friction forces between the grains, due to the increased normal force between adjacent grain surfaces, one would expect that increasing the external pressure on the membranes would have a similar effect. The internal pressure due to the presence of the capillary bridges can be easily evaluated. For ideal spheres, each bridge exerts a force of $F_b = 2\pi R\gamma\cos\theta$ (equation (6)). Roughness can only reduce this force, hence we have in general $F_b \le 2\pi R\gamma\cos\theta$. If $N \approx 6$ is the coordination number of the bridge network and ρ its packing density, we find $p = dE/dV = NF_bR/3V \le N\rho\gamma/2R \approx 1.9\gamma/R$. For beads with a diameter of 200 microns wetted by water, a pressure of $p_0 \approx 7$ mbar is obtained. This sets the scale for the external pressure to apply via the water reservoirs. If the yield stress of the pile was provided by mutual friction between the grains, applying 7 mbar externally

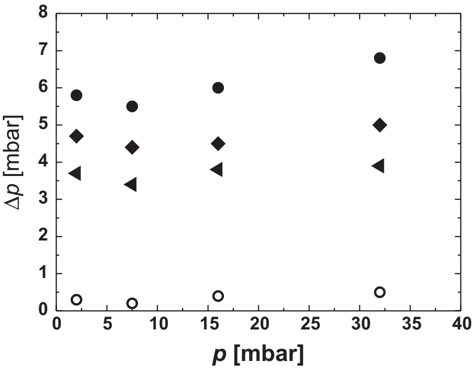

Figure 16. Dependence of the differential pressure Δp on the absolute pressure p for $u = 16.5\,\mu\mathrm{m/sec}$. Open symbols: dry sample. Closed symbols: $W = 3 \times 10^{-3}$ (triangles), $W = 4 \times 10^{-3}$ (squares), $W = 1.4 \times 10^{-2}$ (bullets).

would be expected to increase the measured yield stress to a similar extent as did the addition of the liquid.

The dependence of the yield pressure Δp on the absolute (external) pressure $p := (p_1 + p_2)/2$ is shown in figure 16. It is clearly seen that rising the external pressure even above 30 mbar hardly changes the yield stress of the material. In contrast, the addition of water entails a dramatic change. We see that the effect of the bridges on the yield pressure is much larger than the effect due to increased friction at a pressure corresponding to the internal pressure exerted by the bridges. At first glance, this seems to show that the increase in yield stress of a wet pile of glass beads, as opposed to the dry one, is mainly due to the topology of the bridge network, not to increased friction. However, it must be borne in mind that we are comparing isotropic pressure (as exerted by the liquid bridges) with uniaxial pressure (as applied externally). This may have substantially different effects, in particular on a granular medium. The question posed above can thus not yet be answered.

In what follows, we define the yield pressure as the pressure difference Δp extrapolated linearly to zero external pressure. Its dependence upon the liquid content, W, is shown in figure 17. The upper and the lower graph both show the same measurement, but the lower graph is "zoomed" into smaller W in order to better display variations at very small water content. As it can be seen in the upper graph, the granulate started to become softer for $W > 3\%$. This corresponds to the crossover to regime **IV**, where liquid aggregates are forming. Above the percolation threshold (regime **V**), i.e., for the data points at $W = 0.19$ and $W = 0.27$, experimental scattering in Δp was reported to be quite large. This was possibly due to long distance transport within the percolated liquid structure, which may be very sensitive to the topology of the cluster(s). At full saturation, no interface is present and the granulate is as soft as in the dry case. As a consequence, it could be sheared at nearly no force in the fully saturated case.

From the bottom of figure 17, we see that within experimental scattering, the yield stress scales as the bridge force, which is indicated by the solid line. We shall check if this is what we would expect. Imagine a rectangular sample

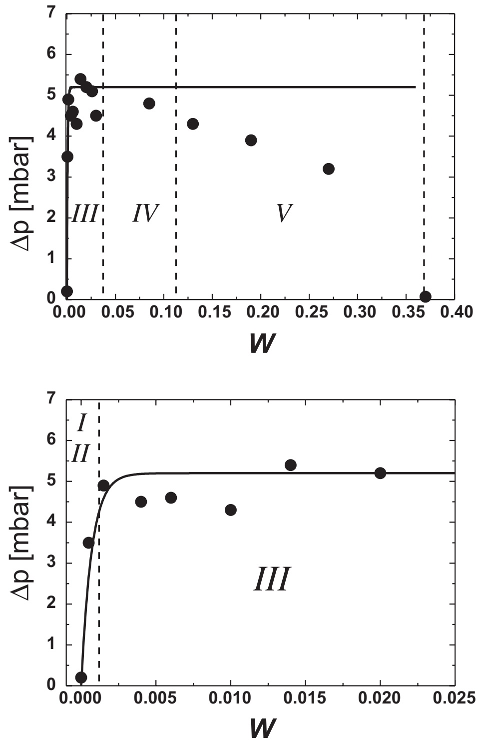

Figure 17. Dependence of the differential pressure Δp on the liquid content W. Top: full scale. Bottom: low water content regime only. The solid line corresponds to equation (13), with V_0 corresponding to $W_0 = 7 \times 10^{-4}$.

of wet granulate, with a volume V_S, which is being subject to a homogeneous uni-axial unity strain by application of an external shear force, F_s. The energy afforded by this force is $F_s \cdot d$, where d is the thickness of the sample, and thus the distance over which the force is to be applied. If there was no friction between the spheres, all of this energy must have been transferred to the capillary bridges. The fraction of such bridges which is ruptured with unity shear may be called $\tilde{G}$, which is a geometric factor of order unity, similar (and certainly related) to the parameter $G \approx 2$ we have discussed in the context of the simulations above. We then have, in the minimal capillary model,

$$F_s d = V_S N \rho \tilde{G} \Delta E_{cap} \tag{35}$$

which can be converted to pressures as $\Delta p \sim F_s d / V_S = N \rho \tilde{G} \Delta E_{cap}$. We see that we would expect the yield pressure not to scale with F_b, but with ΔE_{cap}, which keeps increasing as $W^{1/2}$ in regime **III** (cf. equation (20)). However, we have seen in the simulations (as well as from simple arguments) that G, and very probably also $\tilde{G}$ are decreasing functions of s_c, and thus of W. This may partly cancel, and result in a characteristic indistinguishable from the data within experimental scattering. We should mention that if it was only the friction between the grains, we would

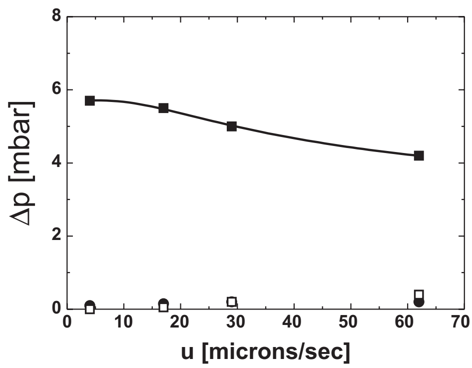

Figure 18. Dependence of the differential pressure Δp on the shearing velocity u, for an absolute pressure of $p = 7.6\,\mathrm{mbar}$. Full circles: dry sample. Open squares: full saturation. Full squares: $W = 0.014$.

just expect the experimental result displayed in figure 17. Again, we have to say that this issue cannot be settled at the present time.

Let us now turn to the dependence of Δp on the shear rate u. As figure 18 shows, Δp decreases significantly as u is increased in the wet, and only in the wet case. This is in qualitative accordance with results obtained before by Tegzes in rotating drum experiments [79]. No clear dependence of Δp on u was found in the dry case. This is not surprising, since the applied absolute pressures were quite small. In this case, neither for the free sliding regime [19] nor for the stick-slip regime [15, 20] a clear dependence of Δp on u [19] is expected. For the underwater case, a small increase of Δp with u for the higher velocities (see figure 18) was observed, which may be due to the viscosity of the surrounding water.

A certain dependence on shear rate is indeed expected on the basis of equation (19). The time a capillary bridge has for formation is inversely proportional to the shear rate, such that at higher shear rate, a smaller bridge volume is expected. In particular, from equation (19) we obtain

$$\Delta p = \Delta_1 + \Delta_2(1 - e^{-u_0/u}) \tag{36}$$

where u_0 should be roughly the sample radius divided by t_0. This reduces both s_c and also F_b, since V_1 is of the same order as V_0. Consequently, no matter what the mechanism of stiffening is, be it by friction or the topology of the capillary network, a smaller yield stress is expected. The functional dependence of $\Delta p(u)$ would be very similar in both cases. With $t_0 \approx$ five minutes, we would expect $u_0 \approx 30\,\mu\mathrm{m/sec}$. The solid curve in figure 18 represents equation (36), with $\Delta_1 \approx \Delta_2$ and $u_0 = 43\,\mu\mathrm{m/sec}$. This is quite reasonable agreement.

Although these results do not allow a distinction between topology and roughness effects, a major conclusion is that the formation dynamics of the liquid bridges can alone account for shear thinning effects. From our above discussion on the capillary bridge network, we know that the relative change of the coordination number, N, is similar to the relative change in packing density. Since dilatancy changes the density only by a few percent, a dramatic effect on N would not be expected. The strong change observed in Δp (up to 20 %) with increasing shear rate can thus not be explained by dilatancy. This is in contrast to a widespread opinion that dilation effects are generally the main reason for shear thinning. This is particularly interesting because shear thinning is a major requirement for avalanching to occur.

It is instructive to compare the behavior displayed in figure 13 with the shear thinning effects shown in figure 18. Identifying the rate of diffusion with the shear rate (i.e., the site exchange rate), and the applied pressure difference with the applied shear force, we observe that there is no shear thinning in the simulation results. In contrast, a larger site exchange rate of the particles (i.e., larger D) corresponds to larger applied stress, F. Although this is at first glance a qualitative discrepancy, it is straightforward to see where it comes from. The shear thinning observed in the experiment was argued to be due to the finite formation time of the liquid bridges. But in the simulation, bridges are assumed to form instantaneously, so this effect must be absent here. Consequently, what we see in the experiment and in the simulation is precisely what we expect on the basis of the respective model.

4.3. *Horizontal agitation: fluidization via energy*

There has been some significant progress recently in experimental studies of the collective statistical dynamics of wet granular systems. Samadani and Kudrolli investigated the influence of liquid on the mixing behavior of systems consisting of large and small spheres [129, 149]. In these experiments, the morphology of heaps formed by pouring mixtures of spherical particles with two different sizes through an orifice was investigated. It was observed that as liquid was added, the tendency to segregate, which is well known for dry granulates and discussed above ('brasil nut effect') [46, 150], was diminished. At a certain wetness, complete mixing occurred.

In order to have a closer look at the observed effects, it would be desirable to perform experiments in which the statistical parameters, such as the granular temperature, are as well defined as possible. In order to achieve this, experiments have recently been carried out in which the sample was fluidized by a rapid circular shaking motion of the cylindrical container about a vertical axis [74]. Effectively, the sample was thus in every horizontal direction bounded by a wall which exerted an oscillating motion of the same well-defined amplitude and frequency. The containers were 300 ml glass bottles which were mounted on a standard test tube shaker. The shaking motion consisted of small horizontal circles with a diameter of 5 mm and a frequency of 20 revolutions per second, which corresponds to a velocity of the container walls of 31.4 cm/sec.

The containers were filled with a mixture of small glass beads with large ones. The radii of the latter were 2.5 millimeters, the radii of the small beads ranged from 0.05 to 0.25 millimeters. In contrast to what one might anticipate on the basis of earlier work [129, 149], the propensity of mixing of the small particles with the large ones did not improve monotonically with the amount of liquid added. Instead, the tendency to segregate became stronger upon addition of water. Only after a certain amount of liquid had been added did the mixing improve.

The behavior at small liquid content can be rationalized in terms of the effective restitution coefficient discussed for the capillary model. As it can be seen from equation (25), ϵ becomes smaller when some liquid is added (because ΔE increases). It is thus expected that effects particular to granular matter, such as the brazil nut effect, become more pronounced, as it was observed.

While at small liquid content the general motion of the whole sample was a kind of circular convection around the axis of the jar, the motion became qualitatively different above a well defined critical water content which depended on the radius of the small spheres [74]. This transition was identified with the one reported by Samadani and Kudrolli. It was argued that the transition occurs when the average kinetic energy per bead is no longer sufficient to break the liquid bridges between the beads. This is a kind of condensation within the granular gas, mediated by the liquid bridges.

In order to support this view, we have to discuss the involved forces in some detail. As it has been shown above, the energy required to destroy a bridge can be written as $\Delta E_{cap} \approx 5.5R^2\gamma\sqrt{\tilde{V}}f_b(\tilde{V})$ (equation (20)). In order to locate the condensation transition, we require the rupture energy to be equal to the average kinetic impact energy of the small spheres, $E_i = m/2\langle v^2\rangle$ [151]. Since $\langle v^2\rangle$ corresponds to the granular temperature, and thus to the irregular part of the particle motion, v is expected to be of the same order, but smaller than the velocity of the container walls [33]. We obtain

$$\tilde{V}R^2 - R^{(2-(1/\mu))}L^{(1/\mu)}\tilde{V}^{(1+(1/2\mu))} + 2\pi\delta^2 = 0 \tag{37}$$

where we have used equation (16) with $\tilde{V}_0 = 2\pi\delta^2/R^2$, and introduced the length scale $L := 2.63\gamma/(\varrho_{SiO}\langle v^2\rangle)$. Equation (37) can be easily solved for $\mu = 1$ and $\mu = 1/2$. Adopting equation (18) for $W(\tilde{V})$, we use $\tilde{V} \approx 2.3\,W$ (cf. figure 5) and obtain for $\mu = 1$

$$R \approx 0.76L\sqrt{W}\left(1 + \sqrt{1 - \left(\frac{W_c}{W}\right)^2}\right) \tag{38}$$

in which $W_c = \sqrt{8\pi}\delta/2.3L$ is a critical water content below which no real solution exists. Equation (38) represents the phase boundary at which condensation is expected to set in, and is depicted as the solid curve in figure 19. From the best fit to the data, one obtains $\delta \sim 500\,\mathrm{nm}$, in agreement with atomic force microscopy of the bead surfaces. As seen from figure 19, it is thus suggested that W_c is of the same order as W_0. From the value of the intrinsic length scale, the average granular velocity, $v := \sqrt{\langle v^2\rangle}$ was found to be $\approx 0.15\,\mathrm{m/s}$. This is of the same order as (but noticeably smaller than) the velocity of the container walls, as expected [152].

For a more pronounced roughness, with $\mu = 1/2$, we obtain

$$R \propto L\sqrt{W - \frac{W_c^2}{W}} \tag{39}$$

which is represented by the dash-dotted line in figure 19. In this case, the good fit shown in the figure is obtained if one assumes that only 40% of the liquid are in the capillary bridges, and the average impact velocity is 10 cm/sec. This is quite

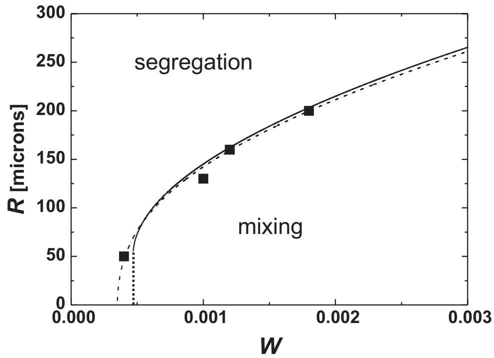

Figure 19. Segregation phase diagram. The squares indicate the liquid content at which a transition to complete mixing is observed. The curves represent the phase boundary between segregation and mixing as expected from the model (solid: $\mu = 1$; dash-dotted: $\mu = 1/2$).

reasonable as well, and demonstrates the robustness of the model with respect to the (usually unknown) character of the roughness.

4.4. *Vertical agitation: fluidization via force*

Another, more classical scheme of fluidization is to oscillate the container vertically. At sufficiently high amplitude and frequency, the grains start to move in an erratic fashion, very much reminiscent of the molecular motion in a liquid, while the density of the granulate remains close to dense packing [29]. In the dry case, it is easy to derive conditions for this type of fluidization to occur. If the vertical position of the container is given by $A \cos \omega t$, where ω is the (angular) frequency of the oscillation, its maximum acceleration is $A\omega^2$. If this is less than the acceleration of gravity, it is clear that the granulate will remain at rest with respect to the container, and no fluidization takes place. If we define $\Gamma := A\omega^2/g$, we can state that $\Gamma > 1$ is a necessary condition for fluidization. Experimentally, one finds that fluidization of dry granulates occurs at $\Gamma_{crit} \approx 1.2$ [17, 29]. It manifests itself by the onset of an irregular, relative motion of the grains, usually accompanied by some convective flow within the container.

This fluidization scheme has been recently applied to a wet granular system [153]. Glass containers with an inner diameter of 26 mm were filled with glass beads to a height of about 30 mm, with average radii ranging from 140 µm to 500 µm. Controlled amounts of liquid were added and the lid closed. The sample was mounted onto an inductive shaker which could apply amplitudes between 1 µm and 1 mm, at frequencies between $\omega = 125$ and $2000\,\mathrm{sec}^{-1}$.

At fixed agitation frequency, the amplitude was increased until a relative motion of the particles was visible through the glass wall. The critical acceleration was found to depend only weakly on frequency. In all cases, a certain hysteresis with respect to the amplitude was observed, i.e., when the amplitude was increased, a higher fluidization threshold was observed than upon reduction of the amplitude. The upper branch was less reproducible than the lower one. This is expected on the

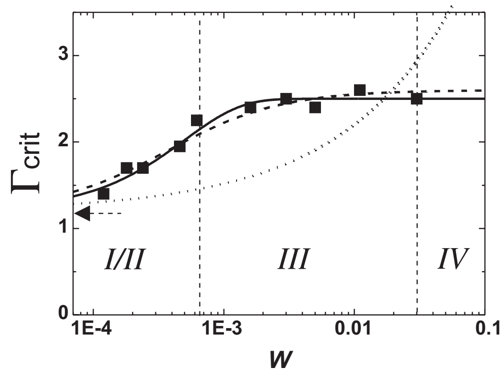

Figure 20. Dependence of the critical acceleration for fluidization on the water content ($\omega = 314\,\mathrm{sec}^{-1}$). Note the logarithmic scale. The arrow indicates $\Gamma_0 = 1.2$. The solid and dashed curve represent equations (13) and (16), respectively. Obviously, no distinction is possible within experimental errors. The dotted line scales as $W^{1/2}$, which certainly does not fit the data.

basis of the aging effects discussed above. When the capillary bridges had some time to develop after the preparation of the sample, they provided larger cohesion than immediately after contact. When fluidization occurred, the sample was always in the dynamic state. In this case there was less liquid for each individual bridge (V_1), but this was rather well defined (equation (19)).

As one would expect, the critical acceleration for fluidization, Γ_{crit}, was found to depend strongly on the water content W, as shown in figure 20. It was found that the data can be well fitted with

$$\Gamma_{crit} = \Gamma_0 + \Gamma_1(R) f_b(W) \tag{40}$$

where R is the radius of the glass beads and $\Gamma_0 = 1.2$ is the critical acceleration for the dry case. In other words, the dependencies on moisture and bead size factorize, at least within the accuracy of the experiments.

Furthermore, this result suggests that the extra acceleration required for fluidizing a wet bed of spherical beads scales with the *force* of the capillary bridges between the beads. This is in contrast to the preceding section, where we have seen that it was instead the *binding energy* (or the energy required to break a bridge) which ruled the fluidization parameters. This is not constant, but scales as $W^{1/2}$, in clear contrast to the data for Γ_{crit} presented here (dotted line in figure 20). This seems similar to the above discussion of the results obtained with the shear cell. However, as it will become clear in a minute, we have here more indications to distinguish between force and energy arguments.

One may argue that since vertical agitation is a scheme based on acceleration, forces should be naturally the paradigm quantities instead of energies. Forces are linked to accelerations just by mass as a pre-factor. We thus have to discuss which mass is the relevant one in our system. In order to understand the dependence of Γ_1 on the bead radius, let us estimate the forces involved in the interaction of the container walls with the glass beads, and the acceleration force acting on the sample. The latter is given by

$$F_{acc} = (\Gamma - \Gamma_0)\varrho_S V_S g \tag{41}$$

where $\varrho_S \approx 1.60 \times 10^3\,\mathrm{kg/m^3}$ is the weight density of the sample, and V_S is the sample volume considered. The capillary forces acting between the container walls and the glass beads are assumed to be similar to the force acting between two beads. The total capillary force is then

$$F_{cap} = F_b \rho_{2D} O_S \tag{42}$$

where ρ_{2D} is the two-dimensional density of the bridges formed between the container wall and the glass beads, and O_S is the total surface considered. For hexagonal close packing, $\rho_{2D} = (2\sqrt{3}R^2)^{-1} \approx 0.29/R^2$. For random packing, ρ_{2D} is certainly smaller than that, but of the same order of magnitude. Relative motion between the glass pile and the container walls will take place if $F_{acc} \geq F_{cap}$. The condition for the onset of fluidization is then

$$\Gamma_{crit} \approx \Gamma_0 + 0.29\cos\theta \frac{2\pi\gamma}{\varrho_S\, g} \frac{O_S}{R\, V_S} \tag{43}$$

It is thus predicted that the extra acceleration due to the addition of moisture, for ideal spheres and high frequency, is inversely proportional to the bead radius. This was indeed found to be the case, at least within the range of radii investigated [112].

So far, the results suggest that fluidization of wet granular matter by vertical agitation, at least for this model system, can be quite well understood by the interaction of the pile with the walls of the container. If this is the case, the fluidization process must depend upon the capillary interaction of the glass beads with the walls of the container. This hypothesis was tested by hydrophobizing the container walls. They were exposed to a solution of octadecyl trichlorosilane, which is well known to bind covalently to the glass surface. As a result, a monomolecular close packed layer of alcane forms on the glass, resulting in contact angles (with water) well above 100 degrees. As discussed above (equations (5) and (6)), there should be no attractive force to the wall in this case. The force between the grains and the wall, which is assumed to mediate the fluidization (equation (42)), should be greatly reduced. In fact, the effect was dramatic: fluidization could not be achieved at all with the apparatus, which was capable of applying accelerations up to $\Gamma \approx 20$.

Finally, let us consider Γ_{crit} in the full range of liquid content, as we have done above for the yield stress [112]. Data averaged over a wide range of frequencies are presented in figure 21. One clearly sees an abrupt change in Γ_{crit} around $W \approx 15\%$. This may be attributed to the percolation threshold, above which there is one cluster of liquid which extends over the full sample. As a consequence, any acceleration Γ will lead to massive transport of liquid within the sample, and thus to transformations of bridge structures and, possibly, reorganization of the bridge network. It is straightforward to imagine that this leads to fluidization, even with $\Gamma \leq 1$. We thus can see that fluidization behavior is very sensitive to the global structure

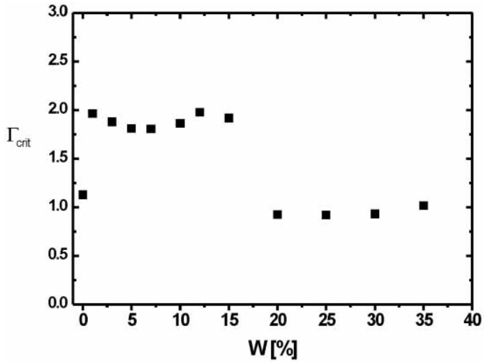

Figure 21. Critical acceleration amplitude for fluidization in a wider range of liquid content. At about 15% water content, a drastic change is observed, leading to fluidization even below $\Gamma = 1$.

of the distribution of liquid within the sample. It may be left to further studies to investigate these effects in detail.

5. Conclusions

There has been considerable progress in recent years in identifying and understanding the microscopic mechanisms underlying the statistical behavior, dynamics, and phase transitions in wet granular materials. As we have seen, a significant fraction of the phenomena observed with this rather complex class of materials can be qualitatively, sometimes quantitatively, understood on the basis of quite simple considerations. There is strong evidence that the hysteretic nature of the capillary bridges forming between adjacent grains is a major ingredient in any successful modelling, while details of the characteristics of the interaction force seem to be of minor significance. The complex behavior of the network of capillary bridges under dynamic conditions has to be investigated in forthcoming studies. In particular, the topology of the liquid interface at higher liquid content and its impact on the mechanical properties of the material remain a challenge for both theorists and experimentalists. In general, the simple model systems discussed here may also serve as paradigmatic systems far from equilibrium, for which there is still a large number of open questions, and some fundamental principles may still await their discovery.

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